

Malabsorption — Celiac Disease





1. HLA-DQ2/DQ8 — negative rules out

WHAT YOU SEE

A 'GENETICS POSTER' with DNA keys 'HLA DQ2' and 'HLA DQ8', 'ALL PATIENTS', 'NEGATIVE RULES OUT', upper-left.

WHAT IT ENCODES

HLA-DQ2 or DQ8 is present in essentially ALL celiac patients, so a NEGATIVE result effectively RULES CELIAC OUT (high negative predictive value). But it is not diagnostic when positive (common in the general population). Useful when serology/biopsy are equivocal or already gluten-free.

2. tTG-IgA (gold star)

WHAT YOU SEE

A detective holding a glowing 'tTG IgA' bottle with gold star, center-left.

WHAT IT ENCODES

First-line serology is tissue transglutaminase IgA (tTG-IgA) — gold star, high sensitivity and specificity. It is the screening test of choice. Endomysial IgA (EMA) is confirmatory/highly specific but more expensive and second-line. Deamidated gliadin peptide is an alternative in young children.

3. Total IgA (catch deficiency)

WHAT YOU SEE

A 'TOTAL IgA' bottle by the detective's leg, lower-left.

WHAT IT ENCODES

Always send TOTAL IgA WITH tTG-IgA — selective IgA deficiency (more common in celiac) causes false-negative IgA serology. If IgA-deficient, use IgG-based tests (tTG-IgG or deamidated gliadin IgG). This pairing is the tested first step.

4. Must be ON gluten

WHAT YOU SEE

Wheat/rye/barley stalks 'WHEAT RYE BARLEY' with an angry gluten figure, center.

WHAT IT ENCODES

The patient must be EATING gluten when tested — testing while already gluten-free causes false-negative serology AND biopsy (mucosa heals). If already gluten-free, perform a gluten challenge (or use HLA typing to exclude). Eliminate wheat, rye, barley; oats often tolerated if uncontaminated.

BOARD TRAP

"Test for celiac while the patient is off gluten" → wrong; a gluten-free diet normalizes serology and histology, giving false negatives. Must be on gluten (or do a gluten challenge).

5. Biopsy: villous atrophy + IELs

WHAT YOU SEE

Flattened mucosa diagram 'FLAT VILLI plus IELS', 'lymphocyte dots', '1-2-3 MARSH', center-low.

WHAT IT ENCODES

Confirm with duodenal (incl. bulb) biopsy: villous atrophy, crypt hyperplasia, and increased intraepithelial lymphocytes (IELs), graded by the Marsh classification (Marsh 3 = atrophy). Take >=4 biopsies. Histology is the diagnostic gold standard in adults.

6. Anti-gliadin (X)

WHAT YOU SEE

An 'ANTI-GLIADIN' tube with a red X, lower-left.

WHAT IT ENCODES

Old anti-gliadin antibodies are obsolete/poor performers and should NOT be the first test. Use tTG-IgA (with total IgA) instead. Deamidated gliadin peptide (a newer, different assay) is acceptable in special cases but classic anti-gliadin is out.

BOARD TRAP

"Order anti-gliadin antibody as the first/screening test" → wrong; classic anti-gliadin is obsolete. First test is tTG-IgA plus total IgA.

7. EMA only (X)

WHAT YOU SEE

An 'EMA ONLY' tube with a red X, lower-left.

WHAT IT ENCODES

Endomysial antibody (EMA) alone is not the first-line SCREEN — it is highly specific confirmatory test, operator-dependent and costlier. Screen with tTG-IgA first, then EMA to confirm if needed. Choosing EMA-only as the initial test is wrong.

BOARD TRAP

"Use EMA alone as the first/screening test" → wrong; EMA is confirmatory. Screen with tTG-IgA + total IgA first.

8. Associations: thyroid, T1DM, DH

WHAT YOU SEE

A bulletin board with 'THYROID', 'T1DM', 'DERMATITIS HERPETIFORMIS', upper-right.

WHAT IT ENCODES

Celiac associates with autoimmune thyroid disease, type 1 diabetes, dermatitis herpetiformis (pruritic vesicles, IgA deposits — treat dapsone + gluten-free), IgA deficiency, Down/Turner syndrome. Screen high-risk groups even if mild symptoms.

9. Acute pancreatitis / leukemia (X)

WHAT YOU SEE

Two boxes 'ACUTE PANCREATITIS' and 'ACUTE LEUKEMIA' each crossed out, upper-right.

WHAT IT ENCODES

Acute pancreatitis and acute leukemia are NOT celiac associations — they are distractors. Celiac's malignancy link is enteropathy-associated T-cell LYMPHOMA (EATL) and small bowel adenocarcinoma, not acute leukemia.

BOARD TRAP

"Acute pancreatitis or acute leukemia is associated with celiac" → wrong; the associations are autoimmune thyroid, T1DM, dermatitis herpetiformis, and the cancer risk is EATL/lymphoma, not acute leukemia.

10. Lifelong gluten-free diet

WHAT YOU SEE

A woman with a basket of food and an empty crossed plate 'GLUTEN-FREE DIET', right.

WHAT IT ENCODES

Treatment is a strict LIFELONG gluten-free diet (no wheat, rye, barley). Adherence heals mucosa, resolves symptoms, and lowers lymphoma/osteoporosis risk. Monitor with tTG-IgA trend. Replete iron, folate, B12, vitamin D, calcium.

11. EATL / osteoporosis / iron

WHAT YOU SEE

A jailed figure 'EATL' with 'OSTEOPOROSIS' and 'IRON' drips, right side.

WHAT IT ENCODES

Complications: enteropathy-associated T-cell lymphoma (EATL) — refractory celiac is a red flag; osteoporosis (calcium/vitD malabsorption); and iron-deficiency anemia, because the PROXIMAL small bowel (duodenum/jejunum) is hit worst where iron, folate, and calcium absorb. Often the presenting clue.